

A SCALAR-DEFECT REDUCTION FOR FINITE WEIL–CCM DICTIONARIES

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ABSTRACT. Recent finite Guinand–Weil dictionaries identify the Galerkin truncations considered by Connes–van Suijlekom and Connes–Consani–Moscovici exactly with zero sums over the nontrivial zeros of the Riemann zeta function. We isolate a different structural fact about how such a finite identity can be forced. Let $Z_{N,L}$ be the concrete $(2N+1) \times (2N+1)$ zeta zero-side matrix of the canonical centered dictionary at aperture $L > 0$, and let $D_{N,L}$ be its deterministic explicit-formula matrix. From explicit-formula agreement only on the coefficient-sum-zero smooth core, we prove that the discrepancy is confined to the two-sided coefficient-sum channel. A second rigidity step, using cancellation of the common nonsmooth tent component in the diagonal basis tests, collapses the entire discrepancy to

$$Z_{N,L} - D_{N,L} = \delta_L J_N, \quad J_N = \mathbf{1}\mathbf{1}^\top,$$

where the scalar $\delta_L \in \mathbb{C}$ is independent of N . Moreover δ_L is exactly the explicit-formula defect of the single literal tent

$$\tau_L(y) = \max\left(0, 1 - \frac{|y|}{L}\right).$$

Consequently the full matrix identity at every finite dimension is equivalent to one scalar identity. A standard C_c^2 mollification argument shows that this defect vanishes and recovers the known exact finite dictionary. The contribution is the scalar-defect reduction and its proof architecture, not a new claim of exact finite zero-sum representability. The core reduction, tent identification, and closure corollaries are formally verified in Lean 4. Nothing in this paper proves the Riemann hypothesis.

1. INTRODUCTION

Weil’s explicit formula admits a positivity formulation equivalent to the Riemann hypothesis, and the associated quadratic functional has been studied from variational, finite-dimensional, and operator-theoretic viewpoints for decades [9, 10, 1, 2]. Recent work has sharpened the finite side of this picture. Connes and van Suijlekom analyze finite-dimensional truncations of real even quadratic forms with the divided-difference structure that appears in the Weil setting [4]. Connes, Consani, and Moscovici restrict the localized Weil quadratic form to the centered $2N+1$ -dimensional Fourier space E_N and use the resulting finite matrices in their zeta spectral-triple program [3]. Groskin then gives an explicit finite Guinand–Weil dictionary: every real even Galerkin coefficient vector determines a band-limited test function whose zero sum equals the finite quadratic value exactly [5].

Accordingly, the statement

$$\text{finite Weil–CCM Galerkin value} = \text{exact sum over nontrivial zeta zeros}$$

is not new here. The question of this paper is narrower and concerns the *proof geometry* of the finite identity:

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How much of the full finite explicit-formula identity must be established before the remaining matrix entries are forced?

For the canonical centered dictionary, the answer is unexpectedly small. Let $Z_{N,L}$ denote the concrete zero-side matrix and $D_{N,L}$ the matrix obtained from the deterministic side of the explicit formula. The individual dictionary basis tests carry one common nonsmooth mode: the literal tent

$$(1.1) \quad \tau_L(y) = \max\left(0, 1 - \frac{|y|}{L}\right).$$

Away from that one mode, coefficient-sum-zero combinations are already smooth enough for the inherited C_c^2 explicit formula.

The first rigidity step is pure finite linear algebra. If a symmetric matrix A is invisible to all pivoted coefficient-sum-zero probes $e_i - e_p$, then

$$(1.2) \quad A = \mathbf{1}a^\top + a\mathbf{1}^\top$$

for an explicit vector a , and therefore $\text{rank } A \leq 2$. For the zero-side/CCM discrepancy $A_{N,L} = Z_{N,L} - D_{N,L}$, smooth diagonal differences then force all diagonal entries of $A_{N,L}$ to agree. Equation (1.2) immediately collapses to

$$(1.3) \quad \boxed{Z_{N,L} - D_{N,L} = \delta_L J_N, \quad J_N = \mathbf{1}\mathbf{1}^\top.}$$

The coefficient δ_L is the centered diagonal discrepancy. Since the centered diagonal basis test is exactly τ_L , one obtains

$$(1.4) \quad \boxed{\delta_L = \sum_{\rho \in \mathcal{Z}} m_\rho \widehat{\tau_L}(\gamma_\rho) - \mathcal{R}(\tau_L),}$$

where $\mathcal{R}$ denotes the deterministic Guinand–Weil right-hand side in the normalization fixed below. In particular, δ_L *does not depend on* N .

Thus an extension problem with $(2N+1)^2$ matrix entries is controlled by one scalar common mode, simultaneously for every finite N . The exact finite matrix identity follows once the single tent defect is shown to vanish. We show that scalar vanishes by a standard smooth mollification of τ_L : the pole and finite prime channels pass to the limit directly, the archimedean channel by dominated convergence, and the complete zero-side sum by Tannery’s theorem (dominated summation).

Main contribution. The principal contribution is the following reduction:

Starting from explicit-formula agreement on the coefficient-sum-zero smooth core, the complete discrepancy between the finite zero side and deterministic matrix is forced into the single N -independent rank-one channel $\delta_L J_N$, with δ_L equal to the literal-tent explicit-formula defect.

The codimension-one completion lemma and the mollified-tent limit are supporting mechanisms. We do not claim priority for the elementary linear-algebra lemma, nor do we claim that approximation of a tent by smooth compactly supported tests is a new analytic technique. We are not aware of the specific N -independent scalar-defect reduction (1.3)–(1.4) in the existing finite Weil literature; we therefore describe it conservatively as an apparently new technical decomposition rather than asserting a global priority claim.

Claim boundary. Nothing in this paper proves the Riemann hypothesis. Exact finite zero-sum dictionaries are already present in current literature [5]; the new point claimed here is the one-scalar extension mechanism. The theorem proves no positivity, negative-inertia persistence, finite-to-infinite spectral convergence, or zero-location statement. The matrices

$$I_N \quad \text{and} \quad J_N = \mathbf{1}\mathbf{1}^\top$$

must not be confused: I_N is the identity matrix, whereas J_N is rank one. The known scalar diagonal normalization of the finite CCM matrix is an I_N -channel statement; the defect theorem of this paper is a J_N -channel statement.

2. FINITE DICTIONARY AND ZERO-SIDE CONVENTIONS

Let $\mathcal{Z}$ denote the set of distinct nontrivial zeros of the Riemann zeta function, with multiplicities $m_\rho \geq 1$, and write

$$(2.1) \quad \gamma_\rho := \frac{\rho - \frac{1}{2}}{i}.$$

We use the Fourier convention

$$(2.2) \quad \hat{f}(z) := \int_{\mathbb{R}} f(y) e^{izy} dy.$$

Fix $L > 0$ and $N \in \mathbb{N}$, where $\mathbb{N} = \{0, 1, 2, \dots\}$. Put

$$\mathcal{I}_N := \{-N, -N+1, \dots, N\}.$$

The canonical centered finite Weil–CCM dictionary supplies symmetric physical-space basis tests

$$k_{nm,L} : \mathbb{R} \rightarrow \mathbb{C}, \quad n, m \in \mathcal{I}_N,$$

with support in $[-L, L]$. In the normalization used here,

$$(2.3) \quad k_{nm,L}(y) = \frac{1}{2} \begin{cases} q_{nm}(|y|, L), & |y| \leq L, \\ 0, & |y| > L, \end{cases}$$

where

$$(2.4) \quad q_{nm}(t, L) = \begin{cases} 2 \left(1 - \frac{t}{L}\right) \cos\left(\frac{2\pi nt}{L}\right), & n = m, \\ \frac{\sin(2\pi nt/L) - \sin(2\pi mt/L)}{\pi(m-n)}, & n \neq m. \end{cases}$$

A coefficient vector $u = (u_n)_{n \in \mathcal{I}_N}$ contracts the basis as

$$(2.5) \quad k_{u,L}(y) = \sum_{n,m \in \mathcal{I}_N} \overline{u_n} u_m k_{nm,L}(y).$$

For real u , put

$$(2.6) \quad s(u) := \sum_{n \in \mathcal{I}_N} u_n.$$

The key regularity decomposition is

$$(2.7) \quad k_{u,L} = s(u)^2 \tau_L + r_{u,L},$$

where $r_{u,L}$ is compactly supported and C^2 for every real u . A direct calculation from (2.3)–(2.4) gives the decomposition; global C^2 regularity of the residual follows by matching the one-sided derivatives at 0 and at the aperture endpoints. This is also one of the formally verified inputs listed in section 8. In particular, $s(u) = 0$ removes the only common nonsmooth tent component.

Let $\mathcal{R}(f)$ denote the deterministic side of the zeta explicit formula in this Fourier convention: the pole contribution, the finite von-Mangoldt contribution, and the archimedean gamma contribution. These deterministic channel expressions are meaningful for the continuous compactly supported basis tests even when an individual basis test lies outside the inherited C_c^2 theorem interface; equality with the zero side is precisely what is not assumed. We define the two finite matrices entrywise by

$$(2.8) \quad (Z_{N,L})_{nm} := \sum_{\rho \in \mathcal{Z}} m_\rho \widehat{k_{nm,L}}(\gamma_\rho),$$

$$(2.9) \quad (D_{N,L})_{nm} := \mathcal{R}(k_{nm,L}).$$

The absolute convergence required in (2.8) is established in proposition 2.1(i). Define the discrepancy

$$(2.10) \quad A_{N,L} := Z_{N,L} - D_{N,L}.$$

Both matrices, and hence $A_{N,L}$, are symmetric in the bilinear transpose sense relevant to the real coefficient contraction.

Choose the centered index $0 \in \mathcal{I}_N$. For $n \in \mathcal{I}_N$ let

$$(2.11) \quad d_n := e_n - e_0,$$

where e_n is the standard coordinate vector. Every d_n has coefficient sum zero.

Proposition 2.1 (Dictionary seam inputs). *For the canonical finite dictionary at $L > 0$ the following hold.*

- (i) *For all $n, m \in \mathcal{I}_N$, the zero-side series defining $(Z_{N,L})_{nm}$ is absolutely summable.*
- (ii) *The polarized physical test corresponding to the pair (d_n, d_m) is compactly supported and C^2 , and the inherited zeta explicit formula applies to it.*
- (iii) *The centered diagonal basis test is the literal tent:*

$$(2.12) \quad k_{00,L} = \tau_L.$$

- (iv) *For every $n \in \mathcal{I}_N$, the diagonal difference*

$$(2.13) \quad k_{nn,L} - k_{00,L}$$

is compactly supported and C^2 , and the inherited explicit formula applies to it.

Proof. The decomposition (2.7) is the common input. Its smooth residual has a Fourier transform with the standard C_c^2 decay, while the literal tent has the quadratic strip decay recorded later in (6.1). Together with the Riemann–von Mangoldt estimate $N(T) = O(T \log T)$, this gives absolute zero-side summability for every real finite dictionary contraction. Polarizing the contractions for $e_n \pm e_m$ then gives entrywise absolute summability, proving (i).

For (ii), the vectors d_n , d_m , and $d_n \pm d_m$ all have coefficient sum zero. Hence (2.7) removes the tent term from every quadratic contraction used in the real polarization identity, leaving compactly supported C^2 residuals to which the inherited explicit formula applies.

For (iii), set $n = m = 0$ in (2.3)–(2.4); the factor $1/2$ cancels the leading factor 2 and gives $k_{00,L} = \tau_L$. Finally, for the coordinate vector e_n one has $s(e_n) = 1$, so (2.7) gives $k_{nn,L} = \tau_L + r_{e_n,L}$. At $n = 0$, (iii) gives $r_{e_0,L} = 0$. Therefore

$$k_{nn,L} - k_{00,L} = r_{e_n,L} \in C_c^2(\mathbb{R}),$$

which proves (iv). □

Remark 2.2. The point of proposition 2.1 is precisely that the individual basis tests need not all lie in the original C_c^2 theorem interface. The nonsmooth part is common. Coefficient-sum-zero polarization and diagonal subtraction remove that common seam before the explicit formula is invoked.

3. CODIMENSION-ONE COMPLETION

We isolate the finite linear algebra independently of zeta.

Let I be a finite nonempty index set and let $A = (A_{ij})_{i,j \in I}$ be a complex symmetric matrix. For real vectors $u, v \in \mathbb{R}^I$ write

$$(3.1) \quad B_A(u, v) := \sum_{i,j \in I} u_i A_{ij} v_j.$$

Fix a pivot $p \in I$ and set

$$d_i := e_i - e_p.$$

Lemma 3.1 (Second differences from pivoted probes). *For all $i, j \in I$,*

$$(3.2) \quad B_A(d_i, d_j) = A_{ij} - A_{ip} - A_{pj} + A_{pp}.$$

Proof. Expand $B_A(e_i - e_p, e_j - e_p)$ bilinearly. □

Proposition 3.2 (Codimension-one matrix completion). *Assume A is symmetric and*

$$(3.3) \quad B_A(d_i, d_j) = 0 \quad (i, j \in I).$$

Define

$$(3.4) \quad a_i := A_{ip} - \frac{1}{2} A_{pp}.$$

Then

$$(3.5) \quad A_{ij} = a_i + a_j \quad (i, j \in I),$$

and therefore

$$(3.6) \quad A = \mathbf{1}a^\top + a\mathbf{1}^\top.$$

In particular,

$$(3.7) \quad \text{rank } A \leq 2.$$

Proof. By lemma 3.1 and (3.3),

$$A_{ij} = A_{ip} + A_{pj} - A_{pp}.$$

Symmetry gives $A_{pj} = A_{jp}$, hence

$$A_{ij} = \left(A_{ip} - \frac{1}{2} A_{pp} \right) + \left(A_{jp} - \frac{1}{2} A_{pp} \right) = a_i + a_j.$$

This is (3.6). Its range is contained in $\text{span}\{\mathbf{1}, a\}$, proving (3.7). □

Corollary 3.3 (Constant diagonal collapses the seam). *Under the hypotheses of proposition 3.2, suppose moreover that*

$$(3.8) \quad A_{ii} = \delta \quad (i \in I)$$

for some $\delta \in \mathbb{C}$. Then

$$(3.9) \quad A = \delta J, \quad J := \mathbf{1}\mathbf{1}^\top.$$

Consequently $\text{rank } A \leq 1$.

Proof. Equation (3.5) with $i = j$ gives $2a_i = A_{ii} = \delta$, so $a_i = \delta/2$ for every i . Substitute back into (3.5). $\square$

Remark 3.4. No novelty claim is attached to proposition 3.2 or corollary 3.3 as isolated linear-algebra statements. Their role is to expose the minimum information needed in the finite Weil dictionary: second-difference agreement plus one diagonal value.

4. SMOOTH-CORE AGREEMENT LOCALIZES THE DISCREPANCY

We apply the completion theorem to $A_{N,L}$ with pivot 0.

Proposition 4.1 (Vanishing on pivoted coefficient-sum-zero probes). *For all $n, m \in \mathcal{I}_N$,*

$$(4.1) \quad B_{A_{N,L}}(d_n, d_m) = 0.$$

Equivalently,

$$(4.2) \quad (A_{N,L})_{nm} - (A_{N,L})_{n0} - (A_{N,L})_{0m} + (A_{N,L})_{00} = 0.$$

Proof. The bilinear contraction in (4.1) is the polarization of finite dictionary contractions by the real vectors d_n and d_m . Since both vectors have coefficient sum zero, the common tent seam cancels. By proposition 2.1(ii), the resulting polarized physical test is C^2 and compactly supported. The inherited explicit formula therefore identifies its complete zero sum with its deterministic right-hand side. Entrywise absolute summability permits the finite polarization to be interchanged with the zero sum. The zero-side polarization is exactly $B_{Z_{N,L}}(d_n, d_m)$ and the deterministic polarization is $B_{D_{N,L}}(d_n, d_m)$, proving (4.1). Equation (4.2) is lemma 3.1. $\square$

Corollary 4.2 (Two-sided coefficient-sum seam). *There exists an explicit vector $a^{(N,L)} \in \mathbb{C}^{\mathcal{I}_N}$ such that*

$$(4.3) \quad A_{N,L} = \mathbf{1}(a^{(N,L)})^\top + a^{(N,L)}\mathbf{1}^\top,$$

where

$$(4.4) \quad a_n^{(N,L)} = (A_{N,L})_{n0} - \frac{1}{2}(A_{N,L})_{00}.$$

In particular,

$$(4.5) \quad \text{rank } A_{N,L} \leq 2.$$

Proof. Apply proposition 3.2 to proposition 4.1. $\square$

At this stage no full finite dictionary identity has been used. The entire unknown matrix has already been reduced to one vector channel.

5. DIAGONAL RIGIDITY AND THE SCALAR-DEFECT THEOREM

The second step uses the common diagonal seam.

Proposition 5.1 (Diagonal rigidity). *For every $n \in \mathcal{I}_N$,*

$$(5.1) \quad (A_{N,L})_{nn} = (A_{N,L})_{00}.$$

Proof. By proposition 2.1(iv), the difference $k_{nn,L} - k_{00,L}$ is a compactly supported C^2 test. Apply the inherited explicit formula to that difference. Linearity of the zero side and deterministic side gives

$$(Z_{N,L})_{nn} - (Z_{N,L})_{00} = (D_{N,L})_{nn} - (D_{N,L})_{00}.$$

Rearrange to obtain (5.1). $\square$

Define the centered defect

$$(5.2) \quad \delta_{N,L} := (A_{N,L})_{00}.$$

By proposition 5.1, every diagonal entry equals $\delta_{N,L}$. The abstract constant-diagonal collapse now applies.

Theorem 5.2 (Scalar-defect reduction). *For every $N \in \mathbb{N}$ and $L > 0$,*

$$(5.3) \quad \boxed{Z_{N,L} - D_{N,L} = \delta_L J_N, \quad J_N = \mathbf{1}\mathbf{1}^\top,}$$

where

$$(5.4) \quad \boxed{\delta_L = \sum_{\rho \in \mathcal{Z}} m_\rho \widehat{\tau_L}(\gamma_\rho) - \mathcal{R}(\tau_L).}$$

In particular, the scalar $\delta_L \in \mathbb{C}$ is independent of the finite dictionary size N .

Proof. By corollary 4.2 and proposition 5.1, the discrepancy satisfies the hypotheses of corollary 3.3. Hence

$$A_{N,L} = \delta_{N,L} J_N.$$

It remains only to identify the centered scalar. By proposition 2.1(iii), $k_{00,L} = \tau_L$. Therefore, from the definitions (2.8)–(2.10),

$$\delta_{N,L} = (A_{N,L})_{00} = \sum_{\rho \in \mathcal{Z}} m_\rho \widehat{\tau_L}(\gamma_\rho) - \mathcal{R}(\tau_L).$$

The right-hand side depends on L and the zeta zero configuration but not on N . Denote it by δ_L . This proves (5.3) and (5.4). $\square$

Corollary 5.3 (Rank-one obstruction). *For every N and $L > 0$,*

$$(5.5) \quad \text{rank}(Z_{N,L} - D_{N,L}) \leq 1.$$

Proof. The range of J_N is the one-dimensional span of $\mathbf{1}$. $\square$

Corollary 5.4 (One scalar controls the full finite identity). *For every N and $L > 0$,*

$$(5.6) \quad Z_{N,L} = D_{N,L} \iff \delta_L = 0.$$

Moreover the same scalar condition $\delta_L = 0$ settles (5.6) simultaneously for every finite N .

Proof. The forward implication follows by evaluating the centered diagonal entry. The reverse implication follows from theorem 5.2. $\square$

Remark 5.5 (Why the reduction is not tautological). If one starts from a complete finite Guinand–Weil dictionary theorem, then $Z_{N,L} = D_{N,L}$ and the discrepancy is already zero. Theorem 5.2 begins from strictly less information: explicit-formula agreement on the coefficient-sum-zero smooth core plus smooth diagonal differences. It identifies the unique obstruction to extending that partial agreement to all finite dictionary entries. The result is therefore best read as a one-dimensional extension principle rather than as a competing proof of the already-known exact finite dictionary.

6. CLOSURE OF THE LITERAL-TENT DEFECT

For completeness we record why the scalar in (5.4) vanishes. This step uses standard approximation machinery and is not the novelty claim of the paper.

The literal tent has Fourier transform

$$(6.1) \quad \widehat{\tau}_L(z) = \begin{cases} \frac{2(1 - \cos(Lz))}{Lz^2}, & z \neq 0, \\ L, & z = 0. \end{cases}$$

In particular it has quadratic decay on horizontal strips of bounded imaginary part.

Choose a real even nonnegative $\varphi \in C_c^\infty(\mathbb{R})$ with $\text{supp } \varphi \subset [-1, 1]$ and $\int \varphi = 1$, and for $0 < \varepsilon \leq 1$ set

$$\varphi_\varepsilon(x) = \varepsilon^{-1} \varphi(x/\varepsilon), \quad \tau_{L,\varepsilon} := \tau_L * \varphi_\varepsilon.$$

Then $\tau_{L,\varepsilon} \in C_c^\infty(\mathbb{R})$, the supports lie in the common envelope $[-(L+1), L+1]$, and

$$(6.2) \quad \widehat{\tau_{L,\varepsilon}}(z) = \widehat{\tau}_L(z) \widehat{\varphi}(\varepsilon z).$$

For every fixed $z \in \mathbb{C}$,

$$(6.3) \quad \widehat{\varphi}(\varepsilon z) \longrightarrow 1 \quad (\varepsilon \downarrow 0).$$

On the closed spectral strip $|\text{Im } z| \leq \frac{1}{2}$, the compact support and normalization of φ give a uniform bound

$$(6.4) \quad |\widehat{\varphi}(\varepsilon z)| \leq C_\varphi \quad (0 < \varepsilon \leq 1),$$

for a fixed constant C_φ .

Proposition 6.1 (Literal-tent explicit formula). *For every $L > 0$, the zero-side family*

$$(6.5) \quad \rho \longmapsto m_\rho \widehat{\tau}_L(\gamma_\rho)$$

is absolutely summable and

$$(6.6) \quad \sum_{\rho \in \mathcal{Z}} m_\rho \widehat{\tau}_L(\gamma_\rho) = \mathcal{R}(\tau_L).$$

Consequently

$$(6.7) \quad \delta_L = 0.$$

Proof. Apply the classical C_c^2 zeta explicit formula to each $\tau_{L,\varepsilon}$. We pass to the limit separately in its four channels.

For the complete zero side, (6.2)–(6.4) dominate the mollified zero summand by a fixed multiple of

$$m_\rho |\widehat{\tau}_L(\gamma_\rho)|,$$

which is summable by the quadratic strip decay of (6.1) together with the Riemann–von Mangoldt bound $N(T) = O(T \log T)$. Dominated summation (Tannery’s theorem) therefore gives convergence to the left-hand side of (6.6).

The pole terms involve only the two fixed frequencies $\pm i/2$, so (6.3) gives their limits immediately. Because all $\tau_{L,\varepsilon}$ are supported in $[-(L+1), L+1]$, the von-Mangoldt channel has one common finite cutoff; after truncating first, its limit is termwise. Finally, (6.1) gives $O((1+t^2)^{-1})$ real-axis decay, while the archimedean gamma density grows only logarithmically. Their product is integrable, and (6.4) supplies a fixed integrable majorant for the mollified family. Dominated convergence passes the archimedean integral to the limit.

The two sides of the smooth explicit formula thus converge respectively to the two sides of (6.6); uniqueness of limits proves the identity. Equation (6.7) is then the definition (5.4). $\square$

Corollary 6.2 (Exact finite dictionary, recovered). *For every $N \in \mathbb{N}$ and $L > 0$,*

$$(6.8) \quad Z_{N,L} = D_{N,L}.$$

Proof. Combine corollary 5.4 with proposition 6.1. □

Remark 6.3 (Novelty boundary for the corollary). Equation (6.8) is not claimed as a new mathematical phenomenon. Groskin proves an exact finite Guinand–Weil dictionary for the Connes–van Suijlekom and Connes–Consani–Moscovici truncations, giving exact zero sums for every finite Galerkin quadratic value [5]. The purpose of corollary 6.2 is to display the consequence of the scalar-defect reduction: one nonsmooth common mode is sufficient to close all finite dimensions at once.

7. RELATION TO EXISTING FINITE WEIL CONSTRUCTIONS

7.1. Bombieri’s finite negative-inertia theory. Bombieri studies the Fourier-transformed Weil quadratic functional and its finite truncations, proving precise negative-inertia statements when off-critical zeros are present [1]. Those truncations are organized by finite subsets of the transformed zero configuration. The centered CCM dictionary considered here is instead indexed by deterministic Fourier modes $-N, \dots, N$. No direct inertia transfer between those two finite objects is used in this paper.

7.2. Connes–van Suijlekom and Connes–Consani–Moscovici. Connes and van Suijlekom analyze the finite-dimensional matrix structure of real even quadratic forms arising from compactly supported distributions, including the divided-difference structure relevant to the Weil setting [4]. Connes, Consani, and Moscovici define finite spaces E_N spanned by centered low-frequency modes and restrict the localized Weil form to those spaces in the zeta spectral-triple construction [3]. These works motivate the finite matrix architecture; the scalar-defect theorem concerns how an exact zero-side realization can be extended from a smooth codimension-one sector.

7.3. Groskin’s finite Guinand–Weil dictionary. Groskin’s finite dictionary theorem is the closest prior result to the endpoint (6.8). It constructs, in closed form, a band-limited Guinand–Weil test from every real even Galerkin coefficient vector and proves that the finite quadratic value is exactly the corresponding complete zero sum [5]. The result here is complementary: it begins with only the smooth coefficient-sum-zero agreement and asks what algebraic freedom remains before the full finite identity is known. Theorem 5.2 answers that the remaining freedom is exactly one scalar common mode, independent of N .

7.4. Companion detector geometry. Two companion manuscripts study different questions. One gives an explicit Gram-determinant negative witness for an isolated transverse off-critical pair block [7]; the other gives complete-population two-translate determinant detectors for hypothetical off-critical zeros [6]. Neither result is used in the scalar-defect proof. In particular, no negative-direction or positivity theorem is imported into the present matrix-extension argument.

8. FORMAL THEOREM CORRESPONDENCE

The theorem-bearing core is formalized in Lean 4 in the public repository

<https://github.com/HMarcusWH/zeta-23-lean>.

The declarations listed below are present on main at commit

317958bb495e2fe603c047480342d24fb0932519.

The toolchain is Lean v4.33.0-rc2 with Mathlib commit

51e6992efd06126df61a496bebf8f49482a4e129.

The zeta function and its standard analytic infrastructure are inherited from Mathlib; formalization of zeta and L -functions in Lean is described by Loeffler and Stoll [8].

Paper result	Lean declaration
Pivoted second-difference identity	<code>codimOneRealPairing_basisDiff</code>
Codimension-one completion	<code>codimOneMatrixCompletion_of_basisDiff</code>
Rank at most two for the seam	<code>rank_codimOneMatrixCompletion_of_basisDiff_le_two</code>
Entrywise zero-side summability	<code>dictionarySpectralMatrix_zero_entry_summable</code>
Zero-side pivoted smooth-core equality	<code>zeroSideMatrix_basisDiff_pairing_eq_smoothCoreZeroPolarization</code>
Discrepancy vanishes on pivoted probes	<code>zeroSideDiscrepancy_basisDiff_pairing_eq_zero</code>
Two-sided discrepancy completion	<code>zeroSideDiscrepancy_eq_completion</code>
Diagonal rigidity	<code>zeroSideDiscrepancy_diagonal_eq_center</code>
Real dictionary tent-plus-residual decomposition	<code>dictionaryTest_ofReal_eq_tent_smul_add_residual</code>
Residual C_c^2 admissibility	<code>dictionaryResidualTest_admissible</code>
Diagonal difference equals the smooth residual	<code>dictionaryDiagonalDifferenceTest_eq_residual</code>
Centered basis test is the literal tent	<code>dictionaryBasisTest_zero_zero_eq_tent</code>
Centered defect equals the tent defect	<code>zeroSideDiscrepancy_center_eq_dictionaryTentDefect</code>
Scalar-defect reduction	<code>zeroSideDiscrepancy_eq_tentDefect_smul_ones</code>
Rank-one discrepancy bound	<code>rank_zeroSideDiscrepancy_le_one</code>
Full matrix equality iff tent defect vanishes	<code>zeroSideMatrix_eq_dictionaryMatrix_iff_tentDefect_eq_zero</code>
Literal-tent explicit formula	<code>dictionaryTent_explicitFormula</code>
Tent defect vanishes	<code>dictionaryTentDefect_eq_zero</code>
Exact finite zero-side bridge	<code>zeroSideMatrix_eq_dictionaryMatrix</code>

TABLE 1. Direct paper-to-Lean correspondence.

The relevant theorem chain was introduced across pull requests #55–#62; its analytic prerequisites were established in the preceding finite-dictionary and smooth-core modules. Each final PR head passed the repository’s four validation checks: the Lean build, RHRC regression suite, R003 normalization audit, and Hermansson formal check. The Lean workflow builds both the complete `Zeta23.CCM` and `Zeta23.ExceptionalZero` umbrellas and rejects `sorry` and explicit project `axiom` declarations in the promoted subtrees. The substantive theorem endpoints report only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`.

The parameter `hs:ZetaSeam` occurring in the concrete zeta declarations is not an RH assumption. It packages multiplicity positivity, reflection symmetry with multiplicity, and local finiteness of the nontrivial zeros. The repository proves the hypothesis-free theorem `zetaSeam:ZetaSeam` from Mathlib’s zeta infrastructure.

Formal claim boundary. The Lean theorem corresponding to theorem 5.2 is a finite matrix identity. It does not assert that the finite matrices are positive semidefinite, does not pass a negative eigenvalue to an infinite operator, and does not prove the critical-line statement. The repository claim registry keeps RH explicitly open.

9. OPEN PROBLEMS

The scalar-defect theorem suggests several extensions that are logically separate from RH itself.

P1. Multiple common seams. The present dictionary has one common nonsmooth mode, and the residual matrix collapses to one scalar times J_N . For a finite dictionary with r common nonsmooth modes, is there a natural extension theorem forcing the residual into an r -dimensional matrix channel, with a rank bound independent of the dictionary size?

P2. Other L -functions. Which automorphic or Dirichlet L -function dictionaries admit the same pattern: smooth agreement on a codimension- r core, cancellation of common boundary modes, and a finite-dimensional defect space independent of the Galerkin cutoff?

P3. Minimal hypotheses. The proof uses only pivoted basis-difference identities and diagonal rigidity. Can the scalar-collapse theorem be characterized abstractly as a universal extension property of a quadratic dictionary, without reference to the explicit form of the CCM kernels?

P4. Quantitative finite detection. The companion two-translate detector theorem produces a strict negative complete Weil direction from a hypothetical off-critical zero, while the present result removes the finite representation ambiguity for the canonical dictionary. A separate quantitative problem is to bound how large a finite centered dictionary must be before it approximates a given negative complete detector strongly enough to force finite negativity. This requires an effective form-domain/core approximation estimate; it is not supplied by the scalar-defect identity.

P5. Proof-compression as a search principle. Here a formally awkward full-dictionary explicit formula was unnecessary for the finite bridge: the discrepancy first collapsed to one common scalar mode, and only that mode required a separate approximation theorem. It is worth testing whether similar “find the defect space first” reductions can compress other explicit-formula or spectral comparison problems.

CONCLUSION

The finite Weil–CCM zero-side comparison appears at first to require a full matrix of nonsmooth explicit-formula identities. In fact the smooth coefficient-sum-zero core already determines all second differences of the discrepancy. Symmetry confines the unknown matrix to a two-sided coefficient-sum channel, and cancellation of the common diagonal tent component collapses that channel further to one scalar multiple of J_N . The scalar is the explicit-formula defect of one literal tent and is independent of N . Thus the extension from the smooth core to every finite dictionary dimension has one obstruction and no more. Showing that obstruction vanishes recovers the known exact finite dictionary; the reduction itself is the structural content of the paper.

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